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Reimagining automotive manufacturing with Generative AI: a technology-organization-environment perspective

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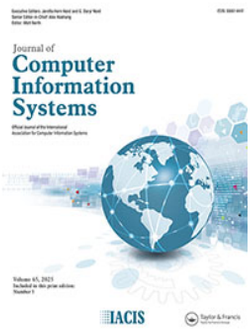
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Reimagining Automotive Manufacturing with Generative AI: A Technology-Organization-Environment Perspective

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ABSTRACT

Generative AI (Gen-AI) is transforming industries, and the automotive sector is no exception. Auto Component Manufacturing (AutoCM) organizations face strong pressure to adopt Gen-AI for competitiveness, yet little research has examined adoption factors in this context. This study investigates the determinants of Gen-AI adoption in AutoCM firms using the Technology-Organization-Environment framework. A model was developed and tested through survey data from 490 managers, validated via PLS-SEM. The findings reveal key predictors of adoption: perceived intelligence, data quality, absorptive capacity, IT infrastructure, top management commitment, vendor support, competitive pressure, and government support. Results further show that adoption intention drives potential use of Gen-AI, but this relationship is weakened by concerns over Gen-AI hallucination. This research identifies crucial drivers of Gen-AI adoption in AutoCM, extends TOE framework literature, and provides actionable insights for practitioners to strategically harness Gen-AI's transformative potential in the sector.

KEYWORDS

Generative AI; TOE; potential use; PLS-SEM; data quality; perceived Gen-AI hallucination potential

Introduction

Gen-AI is a swiftly emerging technology that has reaped significant traction from the industry. Gen-AI creates novel content, including text, code, video, audio, images, and simulations, using machine learning models trained on large datasets.[1,2] While Generative AI, utilizing natural language processing, can create different content formats, such as text and images, its effectiveness relies intensely on the training data's timeliness, quality, and significance. In the Gen-AI ecosystem, although Large language models (LLMs) such as OpenAI ChatGPT, Google Gemini, and Claude are very popular; however, the Gen-AI landscape also encompasses Autoencoders, Diffusion models, Flow models, Generative Adversarial Networks (GANs), Recurrent neural networks (RNNs), Code Llama, and Dall-E, which have widespread business applications in the industry.[3] Generative AI emphasizes synthesis and creating content, images, and videos; unlike traditional AI, which is mainly analytical and predictive in nature. Businesses are using Gen-AI for various functions, including project management, marketing, customer relationship management, content creation, data analysis, human resource management, and coding.[4,5] Worldwide, the market size of Gen-AI is anticipated to grow with a 36.7% CAGR, and the market size will be 136.7 billion in 2030.[3] Furthermore, the 24/7 accessibility of automated Gen-AI has improved efficiency in different business operations.[6,7] Gen-AI reduces manual effort by automating routine tasks and letting employees effectively work on complex and tactical activities. Automation also minimizes hiring costs for routine operations, increasing effectiveness and efficiency in various business functions.[8,9]

Gen-AI has been extensively used in companies such as Synthesia—for video creation, Alphacode, and GitHub Copilot for coding, GPT4 for natural interaction with humans, Dall-E2 for image synthesis and Claude for extensive data processing.[6] Different industry sectors, including insurance, automotive, and media, are exploring applications of Generative AI. The extant research-related Gen-AI adoption has been conducted in the context of marketing,[10] supply chain,[11] restaurants,[12] HRM,[2] and sustainable lean manufacturing.[13]

Furthermore, Gen-AI is extensively used in Auto Component Manufacturing (AutoCM) organizations for research and development, predictive maintenance, supply chain management, manufacturing, quality control, and personalization.[14,15] For example, companies such as Toyota, BMW, and Mercedes use Gen-AI for

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personalization, vehicle assistance systems, and intelligent navigation and routing.[9,16] The AutoCM industry-related Gen-AI market is projected to grow significantly with $2,691.92 million by 2032, at a CAGR of 24.03%.[17] Gen-AI is poised to enhance AutoCM manufacturing operations, resulting in lower maintenance expenses, reduced equipment failures and improved productivity. It is predicted that Gen-AI will lead to a 20% development in equipment accessibility and a potential reduction of up to 10% in annual maintenance costs. Moreover, Gen-AI will provide optimized vehicle performance, personalized driving experiences, comprehensive voice control, and intelligent routing for users of cars.[18]

The Indian auto-component manufacturing sector is diverse, ranging from small and medium-sized enterprises to large multinational corporations. These companies are experts in manufacturing casting, sheet metal fabrication, tools, and dies, and provide services of customized production processes. Most of these companies are small and medium enterprises and face challenges in the adoption of novel technologies.[9,19] However, using Gen-AI technologies can significantly reduce equipment failures, leading to lower maintenance costs and increased productivity of AutoCM firms.[20] There is a rising need for AutoCM firms to adopt Gen-AI to survive the growing threats and stay competitive in the market, as the adoption of AI technologies is still at a nascent stage in AutoCM firms.[15] Pressure from business stakeholders is pushing organizations to accept progressive technologies such as Gen-AI for business competencies, improvement, and an increase in performance.[21–24] Hence, there is a need to understand the various factors influencing the adoption of Gen-AI in AutoCM firms.

Scholars mention that the continued dedication of the research community will drive better advancements in this dynamic arena of Gen-AI.[25] Moreover, the present research does not discuss the acceptance of Gen-AI with respect to the automobile industry and extant research demands this study.[11,26–28] Though Auto-CM companies are planning to adopt Gen-AI, there are concerns related to Gen-AI,[29] as it is a novel technology.[14] Moreover, companies intending to use Gen-AI do not necessarily potentially use it.[30] Various factors influence the adoption of Gen-AI in AutoCM firms. The extant research related to Gen-AI in automotive companies only emphasizes literature related to the use and benefits of Gen-AI[20,31,32] and does not study adoption. Hence, we form the following research question.[9,33,34]

RQ: What factors influence the adoption and potential use of Gen-AI in AutoCM organizations?

This work used the lens of Technology-Organization -Environment[35] to answer this question. This unique research provides the factors influencing the potential adoption of Gen-AI in automobile component manufacturing companies. This work provides valuable insights to companies' managers, marketers, and developers by providing the factors influencing the potential use of Gen-AI. This novel research contributes to academic knowledge by revealing the Gen-AI adoption factors in the automobile industry.

This manuscript includes a literature review and hypotheses. It then details the construct operationalization, research methodology, data analysis procedures, and results. Following this, the discussion section presents managerial and theoretical implications. The paper concludes with suggestions for further exploration and a conclusion section.

Literature review

Gen-AI for AutoCM

Gen-AI comprises Generative Pre-trained Transformer models such as GPT-3 and GPT-4 that utilize large-scale neural networks to generate language based on diverse internet text.[36] One popular Gen-AI application, ChatGPT (OpenAI release in November 2022), gained rapid recognition for its human-like conversational abilities and for seamlessly providing answers to human queries. Gen-AI's language models empower it to create novel content across various formats such as movie scripts, code, e-mails, poetry, reports, and marketing content based on user prompts. Output quality depends on user input and extensive training data.[9,16,37] Gen-AI benefits AutoCM for a) Engineering and Designing: designing processes, developing innovative design options, refining 3D models and optimizing vehicle structure, b) Manufacturing and Production: predicting maintenance needs, efficient production system and optimizing manufacturing processes, c) Autonomous Driving: Creating and testing autonomous driving systems through simulations, developing safe and reliable autonomous vehicles, d) Vehicle image generation: Generation of high-quality images of vehicles for various purposes, such as design, training, and marketing, and e) Marketing and sales: Personalization in marketing and advertising to improve customer experience. Gen-AI also helps generate interactive virtual showrooms to improve customer experience.[9,38]

Gen-AI adoption research

Present Gen-AI adoption research has been conducted to explore the support of top management,[39] generative AI from the perspective of organizations,[28,40]

organizational sustainability,[40] and new product development.[27] Further, research has also explored the adoption of Gen-AI in and Supply chain[41] and online grocery shopping.[42] Also, a study related to the ethics of Gen-AI has been explored.[43] Many scholars are demanding an exploratory investigation related to the adoption of Gen-AI.[4,44–47] The extant research has only a handful of investigations on adopting Gen-AI in an organizational context.[48–50]

Theoretical basis: The TOE framework

Research on technology adoption uses various models to investigate how people accept information systems (IS). The individual-focused models, like the Theory of Reasoned Action,[51] the Technology Acceptance Model,[52] the Theory of Planned Behavior,[53] the Unified Theory of Acceptance and Use of Technology[54] and Innovation Diffusion Theory,[55] primarily concentrate on the user's perspective. These models mostly emphasize technological factors, making these theories and models less suitable for understanding adoption at the organizational level. The TOE framework discusses an organizational viewpoint. It is one of the most dominant frameworks for firm-related technology acceptance,[56] which makes it suitable for this investigation of the acceptance of Gen-AI in AutoCM firms.

The TOE framework identifies and analyzes organizational factors influencing a company's adoption of new technology.[57] The TOE classifies the factors of innovation acceptance into three key facets: technological, organizational, and environmental. The *technological* dimension assesses the relevance of the new technology to the company, considering factors such as complexity, data security and privacy, and compatibility. The *organizational* dimension focuses on the internal resources available to support adoption, including the organization's size, prior technological experience, and potential for cost reduction. Finally, the *environmental* dimension examines external industry characteristics, such as government regulations and competitive pressures, that may impact adoption decisions.[58,59] The TOE framework has been used extensively[60,61] in the acceptance of innovative technologies such as AI in manufacturing,[62] AI-driven industrial robots,[30] AI adoption in the production system[63] and AI in public organizations.[64]

Though TOE is a widely considered model, it has faced limitations and criticism due to its general constructs, and scholars argue that these constructs are insufficient for examining the novel technology adoption, such as AI.[65] To enhance the applicability, the researchers explore novel and contextual factors of novel technology adoption using TOE.[28,65] This work examines the adoption of Gen-AI in auto-component manufacturing companies.

Development of a theoretical framework

The proposed and examined model of this research, depicted in Figure 1, is developed using the TOE framework. The technology-related variables constantly studied are competitive advantages, complexity, and relative and relative advantage.[28] The Gen-AI technology can understand users' requirements and respond to the user with less human intervention. The *perceived intelligence* is the Gen-AI's capability to understand the users' prompts (text and voice) and respond meaningfully and correctly. [66] Auto-CM companies can use Gen-AI to perform product designing, manufacturing, and process optimization activities. Since these tasks are perceived as intelligent; hence, adoption of Gen-AI would happen.[66,67] The scholars highlighted *perceived risk* as a technology-related factor in the context of Gen-AI.[68,69] Gen-AI is subjectively evaluated by users regarding its negative outcomes in AutoCM companies, such as data privacy, copyright concerns, and interaction with various stakeholders using Gen-AI[29,31,69] and over-dependency on Gen-AI results.[69–71] Hence, this research considers perceived risk as a technological factor.[69,72] *Data quality* is related to the accurate, complete, timely, and quality output provided by Gen-AI.[22,65,73] Gen-AI has been used for maintenance prediction, designing, and predictive simulations; therefore, data quality can improve the adoption of Gen-AI.

The organization-related constructs consist of managerial structure, the extent of formalization, firm size, managerial resources, and infrastructure.[28] Organizational antecedents considered from the extant literature in the context of Gen-AI are: Absorptive capacity,[28,74] IT infrastructure,[28,30,75] and top management commitment.[76] The environmental factors adopted from the TOE literature specifically mentioned are related to the industry, competitors, vendors, and government body interactions. This work utilized the environmental factors from the literature, such as Competitive pressure,[30] support from vendors,[63,77] and Government support.[30,78] Moreover, companies that intend to adopt innovative technology, such as Gen-AI, may not use it.[30] Hence, this research investigates the association between adoption intention and the potential use of Gen-AI.

The outputs of Gen-AI are generated using the trained data of a large language model.[79] Gen-AI generates the output from user prompts; however, this

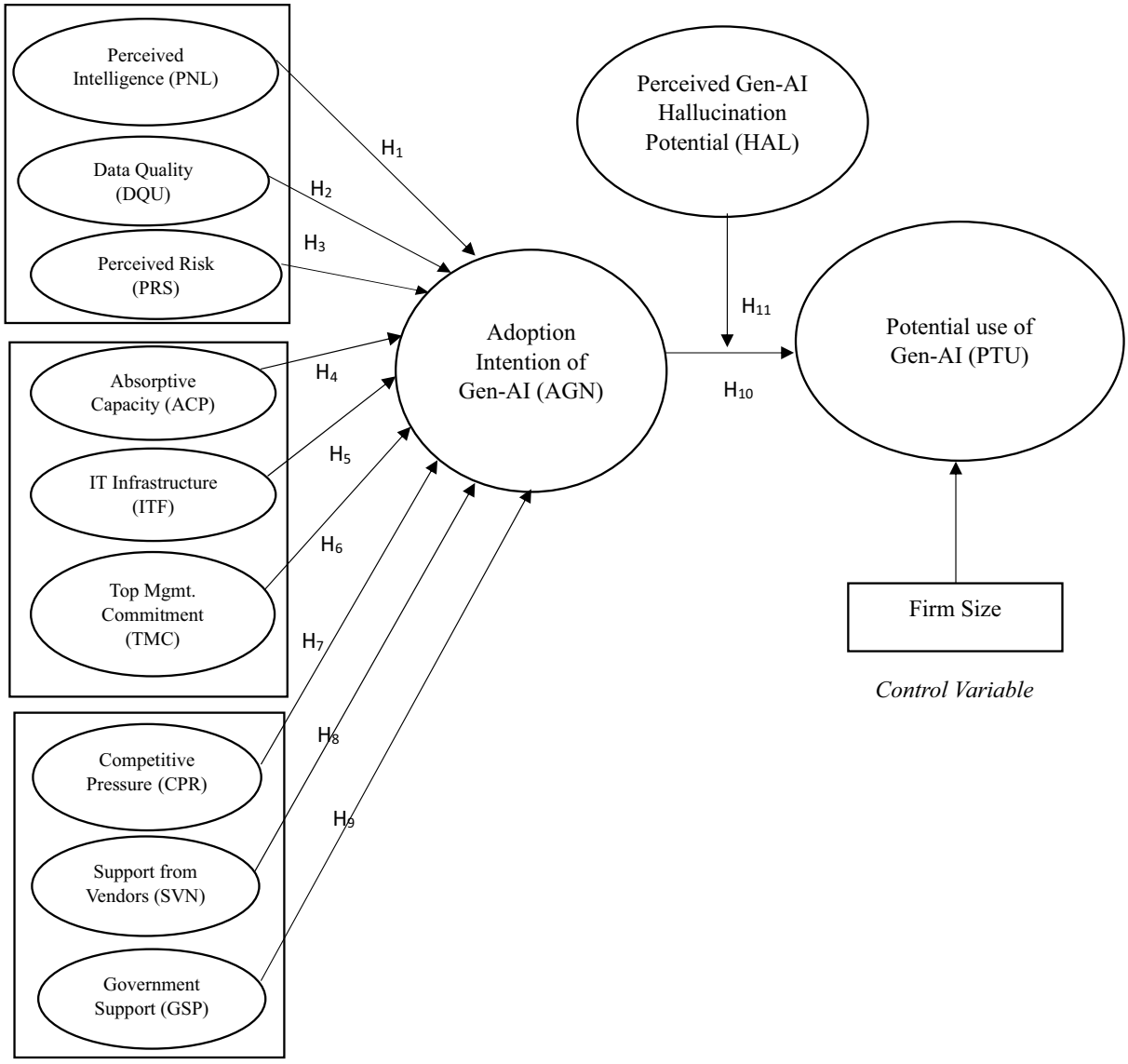


Figure 1. Proposed conceptual framework.

process may provide fabricated and inaccurate information called "AI hallucinations." This issue of hallucinations is a challenge for various business sectors, as accuracy is critical for businesses.[79,80] Gen-AI is used by companies for production, design optimizing, and enhancing features. It includes leveraging virtual prototyping and data-driven design to streamline development processes, personalized in-car assistants for enhanced user experience, predictive maintenance to reduce downtime, and advanced quality control to ensure high standards. However, the hallucinations of Gen-AI might limit its potential usage due to the probability of inaccuracy of results.[80] Hallucinations could result in the spread of false information about autocomponent products or services, potentially damaging their reputation. In the context of AutoCM firms, hallucinations could lead to flawed automobile designs that are not structurally sound or aerodynamically efficient. Hence, Gen-AI might misinterpret sensor data and make incorrect automobile steering, braking, or acceleration decisions.[81] AutoCM firms may intend to use Gen-AI, but increased hallucination potential would make it uncertain to potentially use it. Hence, perceived hallucination potential may negatively moderate the association between adoption intention and the potential use of Gen-AI. The proposed model is presented in Figure 1.

Firm size is included as a control variable, in line with how it's been used in previous information systems studies.[43,82,83] A control variable is treated as a dummy variable. Organizational size is categorized based on the number of employees, distinguishing between small and

medium enterprises, which have between 2 and 999 employees, and large enterprises, which have more than 999 employees.

Hypotheses formulation

The perceived intelligence in chatbots is specifically characterized by their capacity to understand natural language, generate and comprehend responses, and deliver appropriate solutions. This perception of intelligence is a key driver of chatbot adoption.[84] The perceived intelligence of chatbots and robots is discussed as stemming from their ability to demonstrate knowledge, responsible reactions, competence and intelligence,[85,86] a product of AI advancements. Generative AI's capacity to comprehend user requests, engage effectively, and furnish solutions to inquiries[87] affects its adoption by businesses across various functions, especially in sectors such as the AutoCM industry. Hence, we propose:

H_1: Perceived intelligence has a positive association with adoption intention of Gen-AI.

Data quality of advanced technologies is critical for fruitful implementation in organizations as poor data quality can lead to flawed insights, undermining the value of technology investments. Data quality includes the perceived positive attributes of information system output, including completeness, quality, accuracy, timeliness, and availability.[88,89] The present studies related to technology acceptance reveal the association between data quality and technology adoption.[22,89–91] The poor data quality of Gen-AI may be a concern for organizations to implement it.[92] The effective output of Gen-AI in organizations hinges on the quality of its training data. Substandard input data can result in biased, inaccurate, or unreliable outputs.[92,93] The better data quality of Gen-AI outputs would lead to its adoption; hence, the following hypothesis is developed:

H_2: Data quality positively influences adoption intention of Gen-AI.

Perceived risks are multifaceted and individuals may encounter various types of risks, including psychological risks, social, financial and performance.[94,95] Research on various advanced technology contexts supports the idea that perceived risks negatively influence technology adoption.[72,84,95,96] Perceived risk is the judgment and evaluation of the sternness of possible risks related to innovation. In this study, we consider risk as data-related risk, risk related to stakeholder interaction and too much dependency on Gen-AI for various business functions[95] which may put companies into trouble. It was discussed in various studies of Gen-AI acceptance that organizations/people tend to be pessimistic due to perceived risks of Gen-AI use.[72,95,97] Hence, we postulate:

H_3: Perceived risk negatively influences adoption intention of Gen-AI.

Absorptive capacity is an organization's capacity to integrate new external information, recognize its value, and effectively apply it to its commercial objectives. The absorptive capacity is discussed as how organizations acquire knowledge and make efforts to disseminate the same.[98] The extant research reflects absorptive capacity through the perspective of a knowledge repository, mainly focusing on the quantity of preexisting knowledge that organizations possess. An organization's willingness and capability to adopt an innovation are also connected with the absorptive capacity of the organization.[99,100] The research in organizational settings confirmed the association between absorptive capabilities and the adoption of Gen-AI[28] and the adoption of Industry 4.0.[99] The organization tends to have the ability to adopt innovations if they have experience in the use of related and relevant technologies. It is necessary that organizations have the required expertise and technological competencies to accept innovative technologies. Hence, organizations with knowledge and experience related to technologies tend to have the required skills to use and implement Gen-AI.[99] In contrast, firms that do not possess them would not adopt Gen-AI. Therefore, we propose:

H_4: Absorptive capacity positively influences adoption intention of Gen-AI.

IT infrastructure, defined as a shared platform for delivering information with specific range and reach, comprises three elements: physical IT components (*tangible resources*), knowledge assets, customer orientation and synergy *(intangible resources)*, and managerial and technical IT skills *(human resources)*.[101] The Resource-Based View theory admits the significance of resources, which include IT infrastructure.[102] As resource-based theory suggests, tangible resources are key in helping companies swiftly adopt innovations and improve their products. A robust technological infrastructure enables the rapid implementation of new IT applications, giving companies a competitive edge. Hence, tangible resources are necessary because they can affect the adoption of Gen-AI.[28,103] Technical skills are essential for designing, analyzing, and implementing new

business processes, whereas customer orientation is vital for the acceptance of innovations. Customer-focused organizations prioritize innovations that enhance customer satisfaction.

Auto-Component Manufacturing Companies may adopt Gen-AI to accelerate hyper-personalized production and enhance production efficiency and this customer orientation drives Gen-AI acceptance. Knowledge assets, specifically employee skillsets, provide the right prompts to Gen-AI. Companies with rich knowledge bases and skilled workforces are better positioned to adopt and implement innovations. Synergy is achieved when different departments within a company share resources and capabilities, allowing them to respond quickly and flexibly to changing market conditions.[28] Gen-AI represents a revolutionary technology with the potential to fundamentally alter an organization's operative processes and strategic planning. However, harnessing this potential requires substantial IT and management capabilities to adopt Gen-AI. The present literature on AI-driven industrial robots[30] and Gen-AI research confirm the positive influence of IT infrastructure on Gen-AI adoption. Therefore, we formulate:

H_5: IT infrastructure positively influences adoption intention of Gen-AI.

Top management commitment (TMC) represents the degree of dedication and support shown by an organization's senior leadership toward a specific decision.[104] This commitment creates a conducive environment for integrating information technology into business functions. Leadership decisions are crucial when using advanced technology such as Gen-AI.[105,106] The strategic decisions in AutoCM organizations are related to adopting new technology, implementing new technology, new product production, or target markets. TMC is vital for this strategic decision, specifically related to accepting and implementing new Gen-AI technology.[104] The extant literature has established a strong link between TMC and the adoption of AI.[107] TMC promotes recognizing Generative AI's potential, facilitates its integration, addresses resistance to change, and ensures AI benefits are realized. TMC is key to leveraging Gen-AI strategically to improve competitiveness, decision-making and overall AutoCM organization performance. Hence, we propose:

H_6: Top management commitment positively influences adoption intention of Gen-AI.

Within the contemporary, technology-driven manufacturing landscape, AutoCM companies encounter pressure from competitors and technologies such as artificial intelligence and AI assistants for enhanced speed and precision in auto-component production.[78] Gen-AI has been used for customer support, automated inspection, employee performance, support, analysis of sales and information guides for customers.[108] This external pressure is a significant antecedent of adopting innovative technologies.[68] Manufacturing firms, particularly in the Indian market, face pressure to integrate advanced technologies to maintain competitiveness.[109,110] As global automobile companies introduce new vehicle models, AutoCM companies must rapidly respond by delivering customized products.[109] Gen-AI presents a potential solution[28] for Indian AutoCM companies to meet these demands. Therefore, the hypothesis is proposed as below:

H_7: Competitive pressure has a positive association with adoption intention of Gen-AI.

Technology vendor support encompasses the marketing efforts for innovative technology and the training and assistance provided by new technology suppliers. The acceptance of novel technology depends on the marketing efforts of technology vendors. The present research on innovative technology confirms the importance of vendor support and the positive impact of technology acceptance[111] and contradicts too in mobile marketing adoption.[112] Furthermore, employees and management lack awareness of new technologies, necessitating training and support from technology vendors, which can impact the decision to adopt the technology.[113] Gen-AI is a novel and highly advanced technology for AutoCM companies in India, so it would likely require vendor support. Consequently, we propose the hypothesis as follows:

H_8: Support from vendors has a positive association with adoption intention of Gen-AI.

Government support is vital for advancing and promoting emerging technologies, which is also essential for implementing suitable regulations and guidelines to encourage their widespread use.[114] Different government actions, tax compliance, legislation, industrial standards, regulatory measures, and media promotion of emerging technologies are essential.[115] Additionally, governmental policies and protocols contribute to educating businesses and workshops and organizing conferences focused on emerging technological risks.[111] The technology adoption investigations constantly present the importance of government support for technology acceptance in organizations.[30] The Indian

government has implemented several initiatives to boost the auto-component manufacturing sector. The government has established the AutoCM Mission Plan 2016–26 (AMP 2026), which outlines a shared vision for the AutoCM sector. This plan addresses various aspects, including technological growth, contribution to national progress, and global attractiveness.[116] The government provides directives for businesses to seek government approval before implementing specific AI systems in India. It also instructed companies to implement measures to prevent algorithmic bias and the spread of deepfake content.[117] India generally advocates for a regulatory approach to artificial intelligence that encourages innovation. The country aims to maximize AI's potential while addressing the foreseen risks associated with its development and implementation. Hence, we propose:

H_9: Government support has a positive association with adoption intention of Gen-AI.

Companies invest in new technologies when necessary for their business operations.[62,63] The extant research investigated the various stages of the adoption of technology.[58] The Auto-CM companies are planning to adopt Gen-AI for various business purposes, such as optimized product design generation, virtual prototyping, and material selection with data analysis, production planning, and quality control. Though organizations intend to adopt novel technology such as Gen-AI, they may not possess the capability to implement Gen-AI for various business functions.[30,118] The AutoCM companies may intend to adopt Gen-AI technology for various business functions; however, they may not have the capability to implement Gen-AI for potential use. Therefore, the following hypothesis is formulated as follows:

H_{10}: Adoption intention positively influences potential use of Gen-AI.

Gen-AI is developed using machine learning (ML) algorithms to produce new content, including text, images, audio, software code, video, and simulations. The large datasets train the model and generate content as per user prompts.[2] AI software generates text based on user prompts, which can include fabricated information, termed "AI hallucinations."[119] This misinformation poses a significant challenge to companies using Gen-AI for business functions.[80] The scholars discuss the challenges and negative impact on business functions of companies due to hallucinations.[80,119–121] The Gen-AI Hallucinations can provide fabricated, inaccurate, and implausible results to the user. It may affect the business functions of AutoCM companies, such as creating safety hazards in autonomous vehicles due to the misidentification of objects and damaging brand reputation through AI-generated misinformation or flawed product recommendations.[122] Hence, though organizations intend to adopt Gen-AI, it may not potentially end up using it due to the negative impact of hallucinations. Hence, we postulate:

H_{11}: The association between the adoption intention and potential use of Gen-AI is moderated by perceived Gen-AI hallucination potential.

The proposed conceptual framework is presented in Figure 1 below.

Research methodology

This research employed a quantitative methodology. To validate the model, researchers utilized a combination of in-person and online surveys to gather data while overcoming geographical constraints and budget limitations efficiently. The questionnaire covered the study's key constructs and demographic questions about organization size, work experience, education, and previous experience with AI-related technologies. The present literature on TOE was used to design the research instrument to investigate the acceptance of Gen-AI in AutoCM companies. They were adapted from extant literature: Perceived intelligence (PNL),[123,124] data quality (DQU),[22] perceived risk (PRS),[27] absorptive capacity (ACP),[74] IT infrastructure (ITF),[101,125] top management commitment (TMC),[76] competitive pressure (CPR),[126] support from vendors (SVN),[112,127] government support (GSP),[30,78] adoption intention of Gen-AI (AGN),[128] perceived Gen-AI hallucination potential (HAL)[80] and potential use of Gen-AI (PTU).[30,78]

Sampling and data collection

Seven subject experts, including senior officials from AutoCM firms and the Auto Component Manufacturing Association of India, were identified for the study. Before data collection, researchers discussed and explained the research's scope and objectives with these experts. The experts' suggestions were considered and incorporated to confirm face validity, after which a pretest questionnaire was developed. A 5-point Likert scale was employed to measure the variables. The list of AutoCM firms was taken from the ACMA database, which has more than 900 companies in India. The AutoCM firms using artificial intelligence-related

technologies for manufacturing were selected randomly for this research.

To ensure clarity and avoid misinterpretations, the study's questionnaire underwent pre-testing with 20 respondents technology officers, owners, and managers of AutoCM firms), as per the recommendations to refine even standardized questionnaires.[129,130] Minor revisions in the survey instrument were done based on respondent feedback, and its reliability was assessed using Cronbach's alpha. A pilot study with 120 participants was conducted, and the data were analyzed using PLS-SEM with a response rate of 71%. To check the internal consistency reliability of the construct measurement, Cronbach's alpha was calculated, and the coefficient was 0.783. After the satisfactory results in the pilot study, the main data collection was done. The operationalized constructs are demonstrated in Annexure I-Table A1.

The research was conducted using a list of AutoCM firms obtained from the ACMA database. Out of the 950 listed AutoCM firms, 820 were found to be utilizing advanced technology comparable to traditional artificial intelligence in their manufacturing processes and hence were selected for the research. Data sampling focused on automobile hubs in the National Capital region and states such as Tamil Nadu, Maharashtra, Madhya Pradesh, and Gujarat. We used the list of 820 eligible companies and generated random numbers for the sampling process. A random sampling approach was employed to gather data from 490 AutoCM firms from the eligible 820 AutoCM firms across these Automobile hubs. Researchers sought telephone approval before visiting these companies to conduct online or in-person surveys. The study purposively selected individuals from these 490 AutoCM firms who are involved in technology adoption and implementation, such as owners, top management executives, production managers, and technology heads within these organizations.

In the first wave of sampling, from these 490 AutoCM firms, 910 individuals who consented were contacted by presenting the questions on Technological (PNL, DQU, and PRS) and Organizational factors (ACP, ITF, TMC). Out of these 910 questionnaires, we found that 761 responses were appropriate. Furthermore, after a two-week gap, in the second wave, we approached 761 respondents with the list of questions related to Environmental factors (CPR, SVN, and GSP). We found 620 responses deemed to be appropriate and complete. In the third wave, questions related to the adoption intention, perceived Gen-AI hallucination potential, potential use of Gen-AI, and demographic details such as the location of the company, frequency of use of traditional AI technology, and designation of the respondents-related questions were provided. We received 567 responses, out of which 490 were found to be appropriate for final data analysis, eliminating responses with discrepancies such as missing values and incomplete responses. Upon completion of the survey process, 490 out of 910 questionnaires were deemed suitable for analysis, resulting in a 55% response rate. The comprehensive data collection process spanned 14 months between February 2024 and April 2025, with information from diverse AutoCM centers across India.

The survey encompassed 490 AutoCM firms from five regions and states across the country: National Capital Region (23%), Tamil Nadu (21%), Maharashtra (23%), Madhya Pradesh (16%), and Gujarat (19%). Among the surveyed AutoCM firms, 7% have implemented Gen-AI for over six months, while 5% have used it for less than half a year. Additionally, 88% of AutoCM firms employed traditional artificial intelligence for more than a year, alongside other advanced manufacturing technologies. The sample composition includes top management executives (39%), production managers (30%), and technology heads (31%). Furthermore, 71% of the companies employ cutting-edge technologies and 29% use traditional AI for various manufacturing operations.

An independent sample t-test comparing early (310 responses) and late wave (180 responses) data collection groups revealed no significant difference ($p = .36$), indicating the absence of non-response bias.[131,132] A total of 490 complete responses were obtained for the study.

Common method bias (CMB) and endogeneity

A Harman's single-factor test revealed a variance of 26.21%,[133] suggesting that common method bias is not a major concern.[134] This is further supported by multicollinearity VIF values below 5,[135] as shown in Annexure I-Table A1. To address potential endogeneity in the structural model, a Ramsey regression equation error test was performed.[136] Additionally, a marker variable with no conceptual link to the model's constructs, namely Fashion Consciousness (FCS)[137] was used to assess and mitigate method bias,[138–140] further confirming its absence.

Data analysis and results

The steps involved in finalizing the results were the operationalization of the construct measurement, data collection, and data analysis. The data analysis consisted of a data normality check, followed by a measurement model analysis that included convergent validity,

discriminant validity and structural model assessment. Our research utilized PLS-SEM due to its recognized capacity for rigorously evaluating models that contribute to theory development.[141] Additionally, PLS-SEM is well suited for research characterized by complex model structures, predictive objectives, and the incorporation of formatively measured variables.[142] Given our study's extensive sample and analysis of 49 items across 12 constructs, PLS-SEM was deemed the most appropriate analytical approach. This method has been previously employed in various studies examining AI technology acceptance.[143,144] The data's normality was confirmed by Kurtosis and Skewness values falling between '–2' and '+2.' Furthermore, the maximum variance explained by the data was found to be less than 50%.[145]

Measurement model

The measurement model's validity and reliability were assessed using convergent validity, discriminant validity, and reliability tests. Convergent validity was confirmed, as all average variance extracted (AVE) values exceeded the 0.50 threshold.[146,147] Internal consistency, measured by Cronbach's alpha and composite reliability, met the required criteria for all first-order constructs, with Cronbach's alpha exceeding 0.70 for each construct.[148] Discriminant validity, the degree to which measures of different constructs are distinct, was confirmed by comparing the inter-correlations of constructs with the off-diagonal values of average variance extracted, as presented in Annexure I-Table A2. The finding that squared variance values were lower than the corresponding AVE values further supports the discriminant validity between constructs.[149] Further, discriminant validity, assessed using the heterotrait-monotrait ratio (HTMT) of correlations, was also satisfactory, with all values below the 0.85 threshold[150] as mentioned in Annexure I-Table A3.

Structural model

The predictive accuracy of the study's model was evaluated using the coefficient of determination (R^2). The R^2 values for both the adoption intention of Gen-AI (0.636) and the potential use of Gen-AI (0.578) were significant. Adjusted R^2 values were 0.628 and 0.564, respectively. Collinearity was assessed using the variance inflation factor (VIF), with all values within acceptable limits.[151] The model's predictive accuracy using the blindfold method was further confirmed by Stone-Geisser's Q^2 values of 0.706 for the adoption intention of Gen-AI and 0.418 for the potential use of Gen-AI, both exceeding the recommended threshold of zero.[152]

It is revealed that the technology-related factors (presented in Annexure I-Table A4 and Figure A1)- The perceived intelligence ($\beta = 0.306$, $p = 0.002$) and data quality ($\beta = 0.341$, $p = 0.001$) of Gen-AI outcomes influence the AGN, establishing the H_1 and H_2. The perceived risk ($\beta = -0.286$, $p = 0.003$) negatively influences the AGN, confirming H_3. Further, organization-related antecedents—Absorptive capacity ($\beta = 0.216$, $p = 0.004$) of organization and technological resource proficiency ($\beta = 0.233$, $p = 0.004$) positively influence the AGN, supporting H_4 and H_5. The top management commitment ($\beta = 0.272$, $p = 0.003$) is also positively associated with AGN and confirms H_6. Furthermore, the environment-related constructs—competitive pressure ($\beta = 0.346$, $p = 0.001$) influence the AGN, confirming H_7. Also, support from the vendor ($\beta = 0.295$, $p = 0.002$) and government support ($\beta = 0.189$, $p = 0.034$) have positive influences; hence, the H_8 and H_9 are confirmed. Further, AGN ($\beta = 0.288$, $p = 0.002$) positively influences the potential use of Gen-AI and further, this association is negatively moderated by Perceived AI Hallucination Potential ($\beta = -0.196$, $p = 0.038$), which confirms H_{10} and H_{11}. It is revealed that the control variable—firm size does not have any influence on organizational performance ($\beta = 0.026$, $p = 0.001$).

Discussion

This novel study provides an insightful model using the TOE framework for understanding antecedents of the adoption intention of Gen-AI and its influence on the potential use of Generative AI in AutoCM companies. The outcome of this study reveals that the *technology factors*– perceived intelligence influence the adoption intention (H_1), confirming the research on chatbots and robots.[124,153] Gen-AI provides instant outcomes in the form of text, images, and PPTs as per user prompts, along with auto designs and marketing activities campaigns[122] for companies, which portrays that Gen-AI is intelligent.[87] This intelligence of Gen-AI would lead to the adoption of Gen-AI. The data quality provided by Gen-AI can lead to its adoption (H_2), which supports the present research related to blockchain.[89] Gen-AI is developed on large language models (LLMs), creating comprehensive output in a structured manner. It also provides solutions to problems using the most recent data for various business functions.[9] Therefore, data quality leads to better acceptance of Gen-AI in organizations. Perceived risk (H_3) inhibits the acceptance of Gen-AI,[27] which is in line with other technology acceptance which is in line with other technology acceptance research.[72,95,154] AutoCM firms can use Gen-AI for various business functions, where there is

a possibility for cyberattacks, data privacy issues, and copyright issues, which may inhibit organizations from adopting Gen-AI.

Furthermore, the results related to *organizational constructs* are—Absorptive capacity leads to the adoption of Gen-AI in companies (H_4), which is in line with the extant Gen-AI (Agrawal 2024) and other technology[99] research. When the organization needs to use advanced technologies, it tends to motivate employees by exploring and using new technologies such as Gen-AI for various business functions. Also, it was found that IT infrastructure plays an important role in Gen-AI Adoption (H_5), confirming the investigation related to Industry 4.0 technologies.[28,30] Organizations are adopting various IT technologies.[46] IT infrastructure encompasses physical IT resources, employee technical and managerial skills, existing information systems and processes, interdepartmental coordination, and managerial understanding of customer needs. Organizations must develop IT infrastructure and constantly train employees on novel technologies such as Generative AI. Also, companies need to establish the policies, databases, information, and operational procedures for using Generative AI. It was found that top management commitment is a vital predictor of the adoption of Gen-AI (H_6). If the senior executives of the companies are confident with Gen-AI use in various business functions and its capability to provide competitive advantages,[76] they will adopt Gen-AI faster. Moreover, senior management must get involved in the implementation[105] of Gen-AI and ensure the training of staff members for the same.

The results reveal that environmental factors such as competitive pressures are predictors of adoption intention (H_7), which confirms the previous studies of technology adoption[109,110] and Gen-AI solutions.[28] Many AutoCM firms are utilizing Gen-AI for various business functions such as personalized customer communication, optimizing production processes, and improving product efficacy. The AutoCM sector is putting pressure on companies; therefore, it has become imperative for organizations to use Gen-AI.[122] Further, it is found that Support from vendors plays an important role and influences Gen-AI acceptance in companies (H_8).[77,111] Using and implementing Gen-AI in companies requires vendor support, as it is a new technology and organizations need constant support from vendors. Lastly, it is found that government support is the predictor of adoption intention (H_9), which contradicts the research related to industrial robots.[30] The Indian government has developed a National AI strategy for implementation in organizations, promoting the adoption of Gen-AI.[155] Therefore, organizations are in the process of adopting Gen-AI technology. It is revealed that adoption intention is a predictor of potential use of Gen-AI (H_{10})[30,118] and this relationship is negatively moderated by perceived Gen-AI Hallucination Potential (H_{11}), which supports the Gen-AI research in tourism.[80] The organizations intending to use it need to have capabilities. Then, only potentially, AutoCM companies can adopt Gen-AI. Perceived Gen-AI Hallucination potential (HAL) as a moderator has a negative influence on its Potential use in AutoCM companies, as Gen-AI can create imaginary situations by providing instant results that may not be grounded in reality. Also, Gen-AI can be hacked to provide deceptive outcomes. Hence, HAL negatively moderates the relationship between adoption intention and potential use of Gen-AI.

Further, the analysis revealed that firm size as a control variable does not have a statistically significant effect. The lack of significance for firm size as a control variable suggests that the number of employees doesn't necessarily dictate the potential use of Gen-AI in AutoCM organizations. While larger firms with more personnel might utilize Gen-AI extensively, smaller organizations can achieve comparable benefits by strategically applying Gen-AI to specific tasks or areas. Hence, in this study, the control variable firm size does not significantly influence the potential use of Gen-AI in AutoCM firms.

Theoretical implications

Gen-AI has been introduced in the AutoCM Industry and has excellent potential for revamping various business functions. Most research on advanced AI manufacturing technologies, such as Gen-AI, is mainly explored with developed countries instead of developing countries, namely India. The theoretical contributions of this empirical study are discussed as:

This study extends the classical TOE framework by embedding technological dimensions unique to Gen-AI—perceived intelligence and perceived risk operationalizing the Gen-AI specific risk-intelligence dyad within the technology dimension in the AutoCM context. This work provides a novel direction to the TOE framework as it now captures the core affordances and anomalies of Gen-AI, explaining the adoption factors that make it unique as compared to any legacy IT system.

The TOE framework does not inherently provide specific factors for exploring technology adoption. It, however, leverages the framework to categorize and contextualize these factors within the technological, organizational, and environmental domains where Gen-AI adoption occurs. This work advanced the TOE

literature in the context of novel technology such as Gen-AI. The study reveals the antecedents of Gen-AI adoption intention—perceived intelligence, data quality, absorptive capacity, IT infrastructure, Top management support, competitive pressure, support from vendors and government support. Moreover, perceived risk has a negative influence on the adoption intention of Gen-AI.

A multitude of extant TOE studies halt at adoption intention, whereas the model in this study adds potential use as the final outcome, which explicitly positions itself to bridge the intention-behavior gap. The study contributes to quantifying the intention-use gap in the context of Gen-AI in AutoCM firms, offering a richer nomological network among the variables from an organizational perspective.

This work interestingly indicates that though AutoCM firms may intend to utilize Gen-AI, the increased potential for hallucinations would introduce uncertainty regarding its potential implementation. Therefore, perceived hallucination potential negatively moderates the association between adoption intention and potential use of Gen-AI. This study uniquely highlights the negative influence of the perceived Gen-AI Hallucination potential on the potential Gen-AI use, which is not much discussed in recent studies.

This study responds to the scholarly call[11,26–28] for exploring Gen-AI adoption in AutoCM companies. This work enhances the literature on the adoption of Generative AI with an organizational perspective in manufacturing settings. This focus on innovation in manufacturing contributes to a deeper understanding of how new technologies are integrated into industrial processes.

Gen-AI-related factors, such as perceived intelligence and data quality, were unexplored in earlier research. As the Gen-AI systems learn from data, high-quality training data forms the lifeblood of effective Gen-AI systems. These factors are intertwined and crucial for realizing the full potential of generative AI in various AutoCM applications. Moreover, this study contributes to the upper-echelon theory by highlighting the importance of top management commitment in the context of generative AI adoption. Further, organization-related enablers for adopting Gen-AI, such as absorptive capacity and IT infrastructure, are underscored.

Notably, this is a novel work highlighting the influence of adoption intention on the potential use of Generative AI in the context of the automobile sector. Finally, the validated proposed model presents how constructs related to the TOE framework influence Gen-AI's adoption intention and potential use. This validated model has higher explanatory power with adoption intention ($R^2 = 0.628$) and potential use ($R^2 = 0.564$) to provide insightful contributions to the existing body of literature related to Gen-AI. This work enriches the innovative technology acceptance, innovation diffusion literature and makes a significant contribution to the Gen-AI acceptance research with an organizational perspective.

Managerial implications

This work presents the distinctive predictors of the potential use of Gen-AI in AutoCM and offers strategic insights for senior management and IT managers. Moreover, this interesting research has crucial implications for Gen-AI firms and their developers for customizing Gen-AI in the AutoCM context. This research demonstrates how technology, organization, and environmental-related factors influence the adoption intention and potential use of Gen-AI.

The results demonstrate the significance of Gen-AI technology-specific factors, such as perceived intelligence, data quality, and perceived risk, toward its adoption intention and potential use. AutoCM firms' IT managers can collaborate with Gen-AI developers to customize the models in accordance with the context of AutoCM companies' requirements, so that Gen-AI provides intelligent outputs for the users in AutoCM firms. Further, to ensure better data quality, AutoCM firms can direct developers of Gen-AI to train Gen-AI with diverse, domain-relevant, and representative data to ensure the robustness of this training and the data quality of Gen-AI. There should be data cleaning and validation constantly to ensure the quality of data churned through LLM and the outputs of Gen-AI are consistent. AutoCM companies should ensure that the data sources used are ethical, legal, and measures should be taken to eliminate low-quality data. To overcome the perceived risk, the AutoCM companies using Gen-AI for various functions can label the sensitive and confidential data and restrict the input and output of corporate data. As per the various job levels in the AutoCM companies, authentication of Gen-AI data can be provided and confidential data can be encrypted and anonymized. Managers of AutoCM companies can train the employees involved in using sensitive information not to feed the confidential information or trade secrets to the Gen-AI, and also ensure the intervention of humans in the loop while handling sensitive information. Managers can get a service level agreement with Gen-AI firms for not disclosing sensitive data and find trusted Gen-AI service providers in the market. Moreover, Gen-AI companies can ease the perceived risk of using Gen-AI by following ethical standards

and complying with data regulations such as the Digital Personal Data Protection Act 2003 in India.

Moreover, at the organizational level, absorptive capacity, IT infrastructure, and top management commitment play vital roles in adopting Gen-AI. Senior executives and managers involved in adopting Gen-AI must promote Gen-AI use and train employees for Gen-AI usage. As Gen-AI is a fast-evolving and disruptive technology, the AutoCM companies can understand the use of Gen-AI for various functions in their companies and understand how Gen-AI can be integrated with current technology practices. They need to understand how Gen-AI can be helpful for various processes in the company and reap its benefits for the AutoCM firms. The AutoCM senior officials heading the IT teams need to get advice from the Gen-AI companies to ensure the IT infrastructure is upgraded as per the current industry standards. Furthermore, top management needs to understand that their commitment to the adoption of Gen-AI plays a crucial role; hence, they need to strategically implement Gen-AI by understanding how it provides a competitive advantage to the organization.

Environmental factors, such as competitive pressure, vendor support, and government support, are antecedents of Gen-AI adoption. Top management can understand the competitive pressure and adopt Gen-AI in the organization. Top management can benchmark the threshold parameters of Gen-AI used by various competitors to ensure efficiency and effectiveness and become competitive in the market. Managers of AutoCM firms need to have proper service-level agreements with vendors of Gen-AI for their support to ensure the successful acceptance of Gen-AI.

The Indian government has developed a national policy for AI and can take it further to develop the governance framework, especially for AutoCM companies, regarding the use of Gen-AI. The Indian government has already developed a national strategy for AI under the brand #AIFORALL.[156] Moreover, strict data privacy laws and ethical standards can be developed by policymakers to ensure the smooth implementation in companies. Managers of AutoCM and vendors need to understand how the government regulations support the use of advanced AI technology such as Gen-AI.

Further, Gen-AI adoption intention predicts potential use; hence, senior executives need to ensure the capacity to potentially use Gen-AI. Also, perceived Gen-AI Hallucination potential negatively moderates the association between adoption intention and potential use of Gen-AI. As the perceived hallucination potential of Gen-AI increases, the companies may not use Gen-AI, though they intend to. Hence, the Gen-AI and AutoCM companies need to work together to ensure that Gen-AI models are trained on real-time, updated, and accurate data. Senior executives and IT managers need to decipher the hallucinated outputs and resolve them as and when such hallucination issues emerge. Gen-AI companies can implement the Retrieval-Augmented Generation (RAG), which integrates the LLM with knowledge base retrieval systems, which helps Gen-AI to provide trusted and real-time outputs to reduce hallucinations.

Limitations and future research directions

This novel empirical study explores the potential use of Gen-AI in AutoCM companies. Most of the factors revealed in this work would influence various sectors. However, future research could expand this focus to other sectors such as banking, consumer goods, information technology, and services. Further studies can be conducted on AutoCM companies' consumer reactions to the use of Gen-AI technology. Future research can explore the actual use of Gen-AI and its benefits and concerns. Also, studies can be conducted on the ethical use of Gen-AI in various companies. Research can be conducted to explore the contribution of Gen-AI to organizational performance. Even, stages of implementation of Gen-AI and its issues can be explored. This study is cross-sectional and further longitudinal investigation can be carried out. Future research studies on Gen-AI adoption that are exploratory in nature may be conducted and statistical techniques such as post hoc mediation analysis can be carried out. Gen-AI can be used for various functions in AutoCM; hence, experimental research can be done by using it for various functions of Gen-AI. Furthermore, careful consideration needs to be exercised to generalize the findings of this study across various sectors.

Conclusions

Recently, interest and discussions around the applications of Gen-AI across various industry sectors have significantly increased. This study uses the TOE framework to explore the adoption of Gen-AI in the AutoCM industry. As per the findings, the proposed model significantly describes the adoption (63.6%) and potential use (57.8%) of Gen-AI in AutoCM firms. The result revealed the predictors that positively influence Gen-AI adoption, including perceived intelligence, data quality, absorptive capacity, IT infrastructure, top management commitment, competitive pressure, support from vendors and government support. The perceived risk negatively influences of adoption of Gen-AI. Further, the results indicated that the adoption intention of Gen-

AI is an enabler of the potential use of Gen-AI and this association is negatively moderated by perceived Gen-AI Hallucination potential. The results of this research extend the extant literature on Gen-AI acceptance in the automobile industry and provide valuable implications for senior management and managers of companies. Moreover, it provides insights for Gen-AI technology vendors. The study proposed further research in various business sectors using several adoption models.

Author contributions

CRediT: **Brijesh Sivathanu:** Conceptualization, Formal analysis, Methodology, Software, Writing – original draft, Writing – review & editing; **Rajasshrie Pillai:** Formal analysis, Investigation, Methodology, Validation, Writing – original draft, Writing – review & editing; **Nripendra P. Rana:** Conceptualization, Investigation, Supervision, Writing – original draft, Writing – review & editing.

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Annexure

Table A1. Construct measurement and operationalization.

Construct		Factor Loading	AVE	CR	α	Item	Adapted from
Perceived Intelligence (PNL) VIF = 1.286	PNL1	0.826	0.735	0.836	0.798	Gen-AI is competent in solving AutoCM issues related to manufacturing and designing.	123,124
	PNL2	0.878				Gen-AI is knowledgeable in dealing with AutoCM solutions.	
	PNL3	0.864				Gen-AI provides accurate, responsible and correct information	
	PNL4	0.876				Gen-AI used for AutoCM firms is intelligent	
	PNL5	0.842				Gen-AI is functional enough to assist AutoCM firms' issues.	
Data Quality (DQU) VIF = 1.238	DQU1	0.878	0.761	0.842	0.804	Gen-AI is used for providing various information on our company functions.	89
	DQU2	0.854				Gen-AI is utilized for generating comprehensive information.	
	DQU3	0.892				Gen-AI has no errors in the information obtained.	
	DQU4	0.866				Gen-AI information is well-structured and formatted.	
	DQU5	0.854				Gen-AI provides the most recent and current information.	
	DQU6	0.871				Gen-AI provides the most updated and recent solutions.	
Perceived Risk (PRS) VIF = 1.364	PRS1	0.859	0.728	0.873	0.818	There is the issue of data privacy in using Gen-AI	27
	PRS2	0.861				Gen-AI in AutoCM may create data risks hence companies are concerned about its use.	
	PRS3	0.874				Companies are concerned about copyright issues while adopting Gen-AI.	
	PRS4	0.869				Companies are bothered about the interaction of various stakeholders with Gen-AI tools in AutoCM.	
	PRS5	0.892				Companies are concerned that the use of Gen-AI tools may make companies too much dependent on it.	
Absorptive Capacity (ACP) VIF = 1.348	ACP1	0.826	0.725	0.862	0.814	AutoCM company motivates employees to explore Gen-AI in the AutoCM	74
	ACP2	0.884				In AutoCM organization notions and ideas related to advanced technology are discussed at a cross-departmental level.	
	ACP3	0.858				AutoCM company's employees can use advanced technology Gen-AI.	
	ACP4	0.832				AutoCM organization constantly explores the technologies and implements them for various business functions.	
IT Infrastructure (ITF) VIF = 1.326	ITF1	0.822	0.734	0.846	0.808	AutoCM company has the required IT infrastructure available for the adoption of Gen-AI.	101,125
	ITF2	0.839				AutoCM company' staff is well prepared for the knowledge and technical skills required for Gen-AI.	
	ITF3	0.883				AutoCM company anticipates customer necessities in the technology-driven AutoCM industry	
	ITF4	0.861				In AutoCM company, the expertise and capabilities of the workforce are rooted within the established policies, databases, information, and operational procedures.	
	ITF5	0.879				I believe that the different departments of AutoCM company could effectively share Gen-AI resources.	
Top MGT Commitment (TMC) VIF = 1.464	TMC1	0.858	0.739	0.832	0.798	Senior executives are confident that the comprehensive use of Gen-AI will undoubtedly assist AutoCM organizations in effectively improving various functions.	76
	TMC2	0.866				The use of Gen-AI in AutoCM will provide a competitive advantage over rival organizations.	
	TMC3	0.844				In AutoCM organization, managers get involved in the implementation of new technology like Gen-AI for various businesses.	
	TMC4	0.872				Managers in the company provide excellent training in novel technology used in the AutoCM industry.	

(*Continued*)

Table A1. (Continued).

Construct		Factor Loading	AVE	CR	α	Item	Adapted from
Competitive Pressure (CPR) VIF = 1.486	CPR1	0.876	0.726	0.838	0.806	AutoCM company feels the pressure of using Gen-AI due to the adoption of Gen-AI by the competitors.	126
	CPR2	0.872				The use of Gen-AI by the AutoCM industry has forced us to use Gen-AI in our company.	
	CPR3	0.828				AutoCM company monitors rival firms to understand the Gen-AI they employ for various business functions.	
Support from Vendors (SVN) VIF = 1.424	SVN1	0.882	0.755	0.848	0.814	Gen-AI vendors support for implementation in AutoCM.	112,127
	SVN2	0.864				Gen-AI vendors would provide the required training to employees in AutoCM company regarding its use.	
	SVN3	0.898				Gen-AI vendors give free training for the use of Gen-AI.	
	SVN4	0.832				The marketing of Gen-AI is actively done by the vendors of Gen-AI.	
Govt. Support (GSP) VIF = 1.466	GSP1	0.856	0.751	0.836	0.808	Government initiatives are driving the adoption of Gen-AI technology.	30,7830
	GSP2	0.866				Laws and regulations play a vital role in enabling the widespread acceptance of Gen-AI systems.	
	GSP3	0.862				Adequate legal safeguards promote the acceptance of Gen-AI technologies that build upon existing generative models.	
Adoption Intention of Gen-AI in Manufacturing Organizations (AGN) VIF = 1.668	AGN1	0.885	0.755	0.842	0.810	AutoCM company is prepared to allocate resources for implementing Gen-AI.	128
	AGN2	0.873				AutoCM business functions require Gen-AI.	
	AGN3	0.849				Gen-AI is necessary for various business functions in the AutoCM company.	
Perceived Gen-AI Hallucination Potential (HAL) VIF = 1.226	HAL1	0.818	0.709	0.822	0.788	Gen-AI can create imaginary situations that are not grounded in reality, which can create misunderstandings and challenges when attempting to correctly analyze AutoCM function-related information.	80
	HAL2	0.826				Gen-AI is likely to perpetuate cultural biases and stereotypes. This can lead to discriminatory practices and worsen existing social disparities within the AutoCM industry.	
	HAL3	0.846				Gen-AI's dependence on incomplete or flawed data can result in wrong predictions, possibly leading to erroneous assumptions and sub-optimal decisions in the AutoCM industry.	
	HAL4	0.878				Gen-AI is vulnerable to manipulation and hacking; there is a risk of disseminating inaccurate or deceptive information about the AutoCM companies.	
Potential use of Gen-AI (PTU)	PTU1	0.874	0.746	0.832	0.804	AutoCM company plans to use Gen-AI for various business functions.	30,78
	PTU2	0.888				AutoCM company is attracted to using advanced technology such as Gem AI for AutoCM business functions.	
	PTU3	0.862				AutoCM company has the potential to use Gen-AI for different business functions.	

Table A2. Discriminant validity (Fornell and larker 1981).

	TRE	PNL	DQU	PRS	ACP	ITF	TMC	CPR	SVN	GSP	AGN	HAL	PTU
TRE	0.858												
PNL	0.459	0.846											
DQU	0.452	0.455	0.856										
PRS	0.456	0.651	0.567	0.853									
ACP	0.567	0.562	0.621	0.642	0.847								
ITF	0.424	0.481	0.656	0.683	0.679	0.868							
TMC	0.567	0.642	0.535	0.569	0.431	0.678	0.854						
CPR	0.434	0.656	0.672	0.684	0.581	0.632	0.641	0.863					
SVN	0.435	0.632	0.562	0.691	0.613	0.453	0.578	0.651	0.839				
GSP	0.562	0.357	0.465	0.487	0.724	0.678	0.711	0.578	0.672	0.860			
AGN	0.411	0.524	0.673	0.584	0.342	0.456	0.524	0.641	0.601	0.734	0.849		
HAL	0.564	0.467	0.371	0.491	0.361	0.507	0.494	0.573	0.557	0.356	−0.326	0.859	
PTU	0.425	0.245	0.356	.0458	0.678	0.671	0.441	0.641	−.461	0.568	0.491	0.405	0.860

Table A3. Discriminant validity (HTMT).

	TRE	PNL	DQU	PRS	ACP	ITF	TMC	CPR	SVN	GSP	AGN	HAL	PTU
TRE													
PNL	0.597												
DQU	0.506	0.513											
PRS	0.664	0.577	0.575										
ACP	0.548	0.767	0.671	0.676									
ITF	0.535	0.323	0.358	0.453	0.652								
TMC	0.417	0.508	0.538	0.533	0.572	0.659							
CPR	0.528	0.421	0.234	0.378	0.522	0.579	0.599						
SVN	0.407	0.505	0.573	0.569	0.502	0.491	0.302	0.575					
GSP	0.466	0.429	0.471	0.453	0.304	0.756	0.520	0.577	0.446				
AGN	0.475	0.453	0.456	0.459	0.294	0.645	0.578	0.557	0.664	0.568			
HAL	0.562	0.395	0.572	0.481	0.562	0.673	0.781	0.641	0.570	0.571	0.564		
PTU	0.398	0.471	0.562	0.504	0.375	0.562	0.381	0.579	0.397	0.446	0.468	0.427	

Table A4. Structural model assessments.

Hypothesis	Path	β	*t*-statistics
H_1	PNL→ AGN	0.306***	4.984
H_2	DQU→AGN	0.341***	5.317
H_3	PRS → AGN	−0.286***	3.466
H_4	ACP→ AGN	0.216***	2.784
H_5	ITF→AGN	0.233***	2.973
H_6	TMC → AGN	0.272***	3.019
H_7	CPR→ AGN	0.346***	5.898
H_8	SVN→ AGN	0.295***	4.085
H_9	GSP→ AGN	0.189**	2.368
H_{10}	AGN→ PTU	0.288***	3.994
H_{11}	AGN → HAL→ PTU	−0.196**	2.057

[**Note**: t-values for two-tailed test: ***t-value 2.58 (sig. level:1%), **t-value 1.96 (sig. level: 5%), *t-value 1.65 (sig. level: 10%)] **Acronyms**: Perceived Intelligence (PNL), Data Quality (DQU, Perceived Risk (PRS), Absorptive Capacity (ACP), IT Infrastructure (ITF), Top MGT Commitment (TMC), Competitive Pressure (CPR), Support from Vendors (SVN), Government Support (GSP), Government Support (GSP), Adoption Intention of Gen-AI (AGN), Perceived Gen-AI Hallucination Potential (HAL), Potential use of Gen-AI (PTU)].

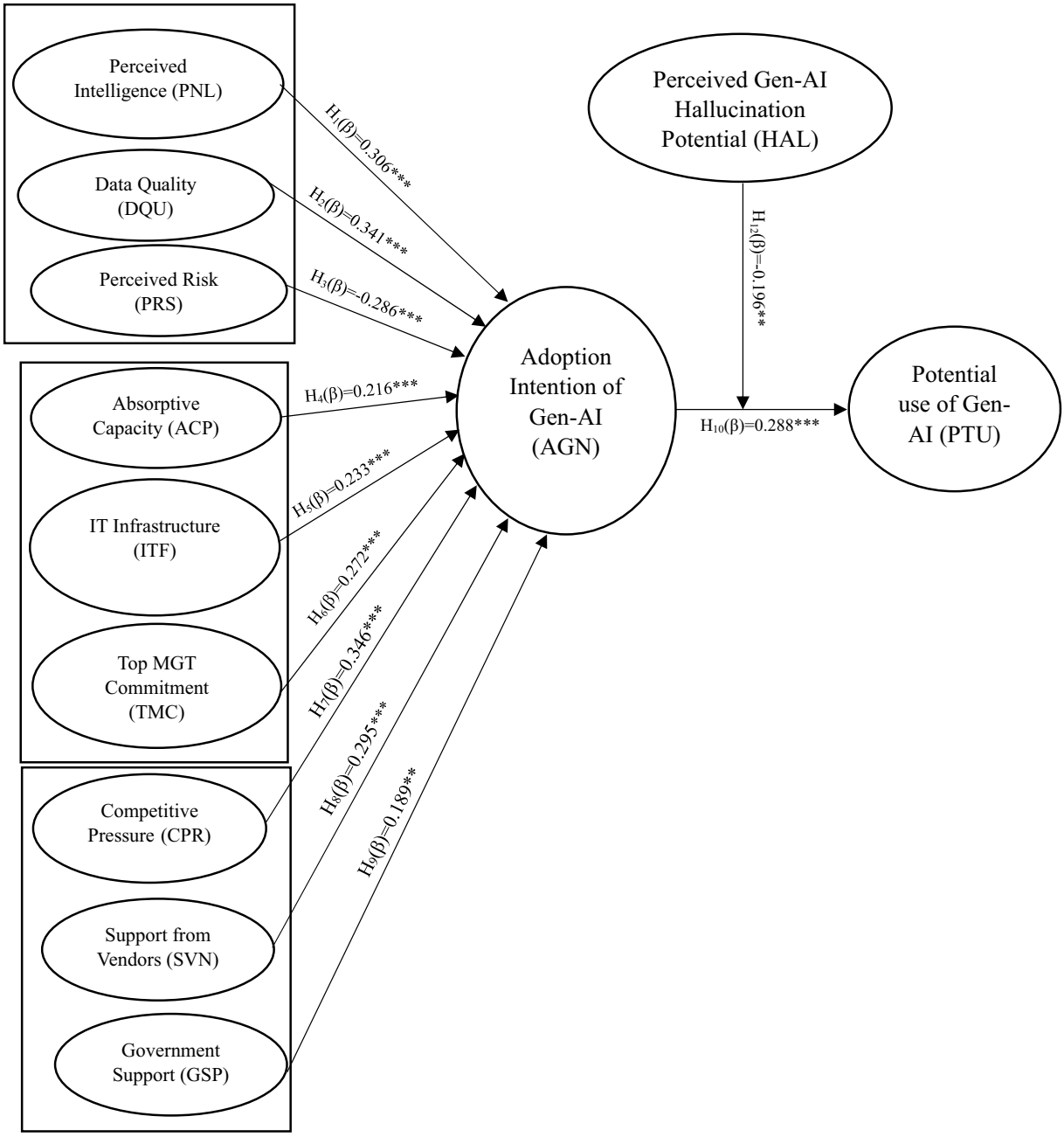


Figure A1. Validate research model.